SIDDHANT PRADHAN

siddhantpradhan722@gmail.com | Amherst, MA 01003
(413) 466-1509 | linkedin.com/in/pradhan-siddhant22

SUMMARY

Computer Engineering senior with experience designing embedded systems that bridge hardware and software, developed through coursework and internship work. Proven collaborator with strong communication skills built through research and campus roles.

EDUCATION

University of Massachusetts Amherst Anticipated Spring 2027
Candidate for Bachelor of Science, Computer Engineering, **Minor:** Engineering Management, **GPA:** 3.67
Awards: Chancellor's Award
Relevant coursework: Embedded Systems, Systems Programming, Hardware Organization & Digital Design, Security Engineering

PROJECTS

Video Capture and Software Image Processing on DE1-SoC Spring 2026
- Built a real-time image processing system on the DE1-SoC, integrating an ARM-controlled video pipeline with the FPGA video subsystem to stream, freeze, and process 320x240 RGB565 frames from a D5M camera module on a VGA monitor
- Implemented memory-mapped I/O from Linux user space using mmap() and /dev/mem and wrote four image processing routines including vertical flip, horizontal flip, black-and-white threshold, and pixel inversion in C

Nios II Embedded System Design on DE1-SoC Spring 2026
- Designed and deployed a hardware-software embedded system on a Cyclone V FPGA, integrating a Nios II soft-core processor with SDRAM, parallel I/O peripherals, and an interval timer configured in Quartus Platform Designer
- Implemented edge-triggered interrupts, a custom decimal display function in C, and deployed the application from Ubuntu Linux on the ARM HPS via scp and shell tooling

Multithreaded Data Acquisition and Synchronization Spring 2025
- Developed a multithreaded embedded application on a BeagleBone Black to capture, coordinate, and store high-frequency hardware events in real time
- Applied thread synchronization, shared data structures, and persistent storage to ensure safe access, correct ordering, and accurate real-time data processing

Interrupt-Driven PWM Timestamping System Spring 2025
- Developed an event-driven embedded system on a BeagleBone Black that captured externally generated signals and recorded timing data using GPIO interrupts without continuous polling
- Coordinated interrupt handling, multithreaded execution and peripheral configuration to achieve reliable synchronization between physical signals and software response

WORK EXPERIENCE

Baxter International Inc., Round Lake, IL Summer 2026
R&D - Hardware and Cyber Engineering Intern
- Built a two-factor access control prototype for an infusion pump lockbox on an ESP32, coordinating an NFC reader, keypad, and linear actuator latch into a single authentication workflow with timeout handling and timestamped event logging

Human Performance Laboratory, University of Massachusetts Amherst Spring 2026 – Present
Undergraduate Research Assistant
- Developing virtual reality experiments to study truck drivers' responses to automated vehicle technology by integrating VR hardware with experimental software, troubleshooting technical issues, and supporting data collection

Minds With Purpose Foundation, Boston, MA Spring 2026
AI Software Engineer Fellow
- Built user-facing features and scalable UI architecture for a social accountability platform using modern web frameworks, resulting in streamlined navigation and measurable increases in community engagement through strategic UX improvements

LEADERSHIP

Executive Board Member of the ECE Student Advisory Council (ESAC) Spring 2025 – Present
- Serving as a liaison between ECE students and faculty, addressing concerns and refining courses based on student feedback
- Assisting in creating a comprehensive semester-based ECE course guide to support students with course and elective selection

SKILLS AND INTERESTS

Skills: C, Python, Bash/Shell Scripting, Unix/Linux Systems Programming, Git, ARMv8, ESP32, Raspberry Pi, BeagleBone Black, Altera DE1-SoC, GPIO/SPI/I2C/UART Interfaces, Memory-Mapped I/O, Multithreading, Peripheral Interfaces, Test Equipment
Interests: Guitar, Cricket, Soccer, Public Speaking, LEGO building